

Connect two sites technique

- **Packet switched** same as Frame Relay
 - Dynamic bandwidth allocation
 - Error checking and retransmission
- **Circuit switched** but it's used to **transport voice not data**
 - Guaranteed bandwidth
 - no error checking
 - retransmission
 - expensive
- **Leased line as a reserved link**
 - Reserved bandwidth
 - high reliability
 - Very expensive



VPN benefits

- **Cost Savings** (no need for **ISPs service Fees**)
- **Security** (carries information within a **private network over public network**)
- **Scalability** (Remote Access VPN)
- **Compatibility of WAN** link options (supports **all popular broadband technologies**)

Key threats to WANs and Remote Access

- The **key threats** to data privacy:
 - Eavesdropping attack
 - Masquerading attack
 - Man in the middle attack

Eavesdropping Attack

- A packet sniffer is a **software application** that uses a **network adapter card** in **promiscuous mode** to **capture all network packets**.
- Packet sniffers **exploit information passed in clear text**.
 - **Protocols that pass information in the clear include the following:** Telnet, FTP, SNMP, POP, HTTP.
- **Common solutions for companies to protect information are:** HTTP with SSL (HTTPS) and implementing a Virtual Private Network (VPN) with encryption

Masquerading attack

- A masquerading attack is where an individual **hides their identity, possibly even assuming someone else's identity**, example: IP spoofing.
- **IP spoofing** occurs when a hacker **inside or outside** a network **impersonates the conversations of a trusted computer**.
- Two general techniques are used during **IP spoofing**:
 - A hacker uses an **IP address** that is **within the range of trusted IP addresses**.
 - A hacker uses an **authorized external IP address that is trusted**.
- The most common solution is to use a **packet integrity** check system, which is implemented with a **hashing function**.

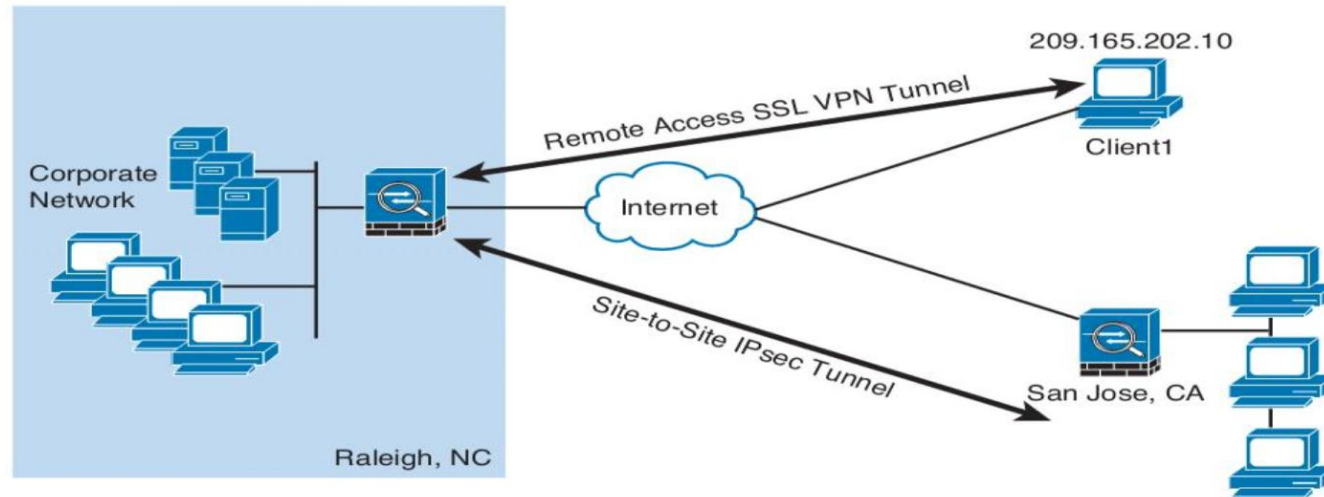
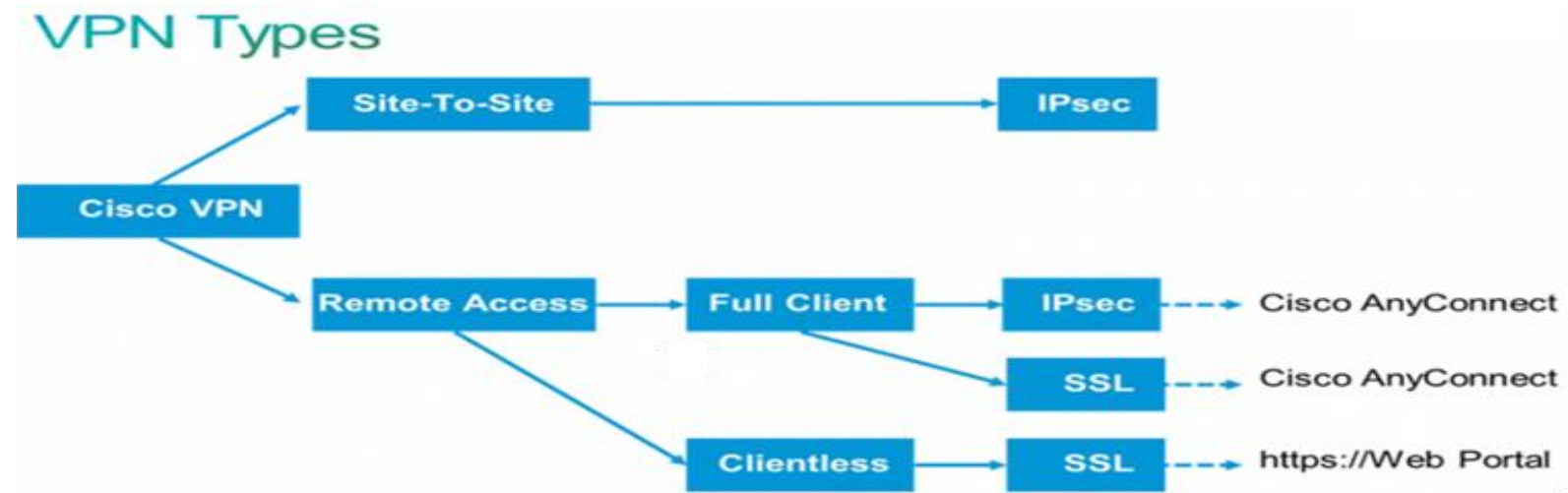
Man-in-the-Middle

- requires that the **hacker** have **access to network packets** that come **across a network**.
- implemented using the following:
 - Network packet sniffers
 - Routing and transport protocols
- Several **solutions** for man-in-the-middle attacks are:
 - A firewall system **randomizes TCP sequence numbers**
 - **VPNs** provide **three tools** to combat man-in-the-middle attacks:
 - device authentication
 - packet integrity checking
 - and encryption

VPN Topologies

- Site-to-site VPNs.
 - Two or more sites that they want to connect securely together (likely using the Internet).
- Remote-access VPNs.
 - Users that build a VPN connection from their individual computer to the corporate headquarters

VPN topology



VPNs remote access

- IPSec VPN
 - Implements security of **IP packets at L3**, and can be used for **site-to-site VPNs** and **remote-access VPNs**.
- SSL (Secure Sockets Layer) VPN
 - Implements security of TCP sessions over encrypted SSL tunnels.
 - Can be used for **remote-access VPNs** (also used to **securely visit a web server** that supports it via HTTPS).

VPN remote access

Feature	IPsec	SSL
Applications supported	Supports All Apps	web-based applications
Authentication strength	Strong	Moderate
Encryption strength	Strong	Moderate – Strong
Connection complexity	Moderate	Low
Connection option	Only specific devices with specific configurations	Any device with a web browser

IP Security Overview

1. In 1994, the **Internet Architecture Board (IAB)** issued a report titled “**Security in the Internet Architecture**” (RFC 1636).
2. The need to **secure the network infrastructure** from **unauthorized monitoring** and **control of network traffic** and the **need to secure end-user-to-end-user traffic** using **authentication and encryption mechanisms**.
3. To provide security, the **IAB** included **authentication and encryption** as **necessary security features** in the **next-generation IP**, which has been issued as IPv6.
4. Fortunately, these **security capabilities** were **designed** to be **usable both** with the **current IPv4** and the **future IPv6**.

Applications of IPsec

1. **IPsec** provides the capability to **secure communications** across a **LAN**, across **private and public WANs**, and across the **Internet**.

Examples:

- **Secure branch office connectivity** over the Internet: A company can **build** a secure **VPN** over the Internet or **over a public WAN**.
- This enables a business to **rely heavily on the Internet** and **reduce its need for private networks**, saving costs and network management overhead.

Applications of IPsec

- **Secure remote access over the Internet:** An **end user** whose system is equipped with **IPsec protocols** can make a **local call to an Internet Service Provider (ISP)** and **gain secure access to a company network**. This **reduces the cost** of toll charges for traveling employees and telecommuters.
- **Establishing extranet and intranet connectivity with partners:** IPsec can be used to **secure communication** with other organizations, ensuring **authentication and confidentiality** and providing a **key exchange** mechanism.

Applications of IPsec

- **Enhancing electronic commerce security:** Even though some Web and electronic commerce applications have built-in security protocols, the use of IPsec enhances that security.
- IPsec guarantees that **all traffic designated by the network administrator is both encrypted and authenticated**, adding an **additional layer of security** to whatever is provided at the **application layer**

Applications of IPsec

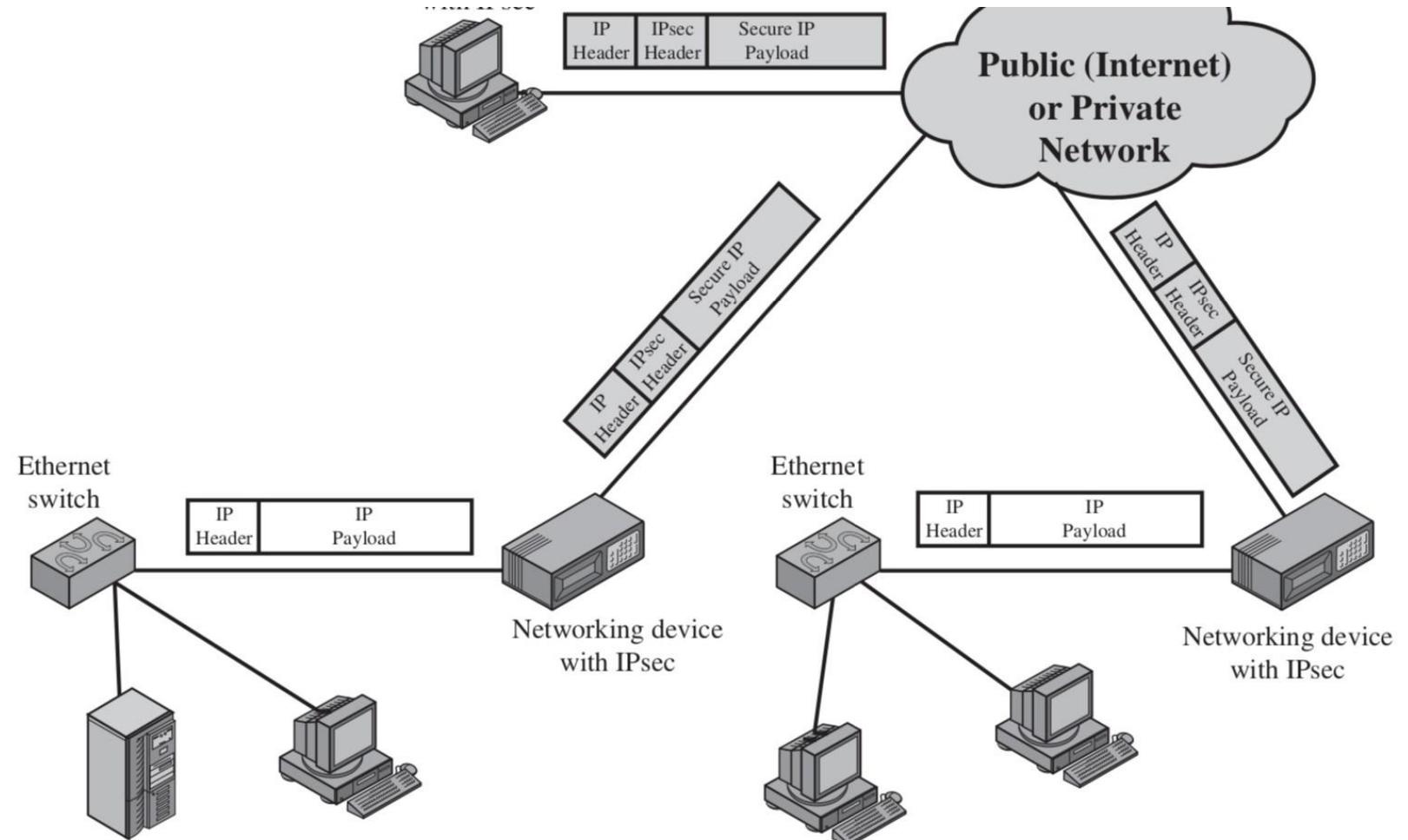


Figure 19.1 An IP Security Scenario

Benefits of IPsec

- When IPsec is implemented in a **firewall or router**, it provides **strong security** that can be applied to all traffic crossing the perimeter. **Traffic within a company or workgroup** does not incur the overhead of security-related processing.
- IPsec in a firewall is **resistant to bypass** if all traffic from the outside **must use IP** and the **firewall is the only** means of entrance from the Internet into the organization.

Benefits of IPsec

- IPsec is **below the transport layer (TCP, UDP)** and so is **transparent** to applications. There is **no need to change software** on a user or server system when IPsec is implemented in the firewall or router. Even if IPsec is implemented in end systems, **upper-layer software**, including applications, **is not affected**.
- IPsec can be **transparent to end users**. There is **no need to train users on security mechanisms**.
- IPsec can provide security for **individual users** if needed. This is useful for **offsite workers** and for setting up a secure **virtual subnetwork** within an organization for **sensitive applications**.

IPsec

- IPsec A **collection** of protocols and algorithms used to protect IP packets at Layer 3
- **Provide:**
 - **Encryption** algorithms for confidentiality: DES, 3DES, AES.
 - **Hashing algorithms** for integrity: MD5, SHA.
 - **Authentication algorithms:** Pre-shared keys (PSK), RSA digital signatures.
 - **Key management:** for ex. Diffie-Hellman (DH), which can dynamically generate symmetrical session keys

IPSec services

- **Authentication Header (AH):**
- **Protocol:** IP Protocol number 51.
- **Services:** Primarily provides **data integrity, data origin authentication, and anti-replay protection.**
- **Mechanism:** AH **calculates a hash** (or "integrity check value" - ICV) over almost the **entire IP packet** (including some **immutable fields** of the **IP header and the entire payload**). **This hash is then included in the AH header.** The receiver recalculates the hash and compares it.
- **Does NOT provide Confidentiality (Encryption):** The data itself remains in plaintext. *AH is rarely used alone in modern VPNs over the internet due to this lack of encryption.*

IPSec services

- **IPSec** Encapsulating Security Payload (ESP)
- **Protocol**: IP Protocol work on **port 50**.
- **Services**: Provides **confidentiality (encryption)**, and optionally **data integrity, data origin, authentication, and anti-replay protection**. It's the most widely used IPsec protocol.
- **Mechanism**: ESP *encrypts the payload of the IP packet*. It can also **calculate a hash** over the encrypted payload (and sometimes the ESP header) for integrity. *The ESP header is inserted between the original IP header and the payload*, and an ESP trailer is appended after the payload.

IPsec services



Connectionless integrity

ضمان سلامة البيانات دون الحاجة إلى اتصال مستمر



Access control

التحكم في الوصول إلى الموارد والبيانات



Rejection of replayed packets

منع إعادة إرسال الحزم القديمة (جزء من سلامة الاتصال)



Data origin authentication

التحقق من مصدر البيانات



Limited traffic flow confidentiality

حماية محدودة لسرية تدفق البيانات



Confidentiality (encryption)

حماية سرية البيانات باستخدام التشفير

SSL

- There is **not an IPsec client** or software currently running on **everybody's** computer.
- Even if there were, **not everyone** has a **certificate** or a **PSK** for authentication.
- Every **web browser** on **every** computer supports **SSL**

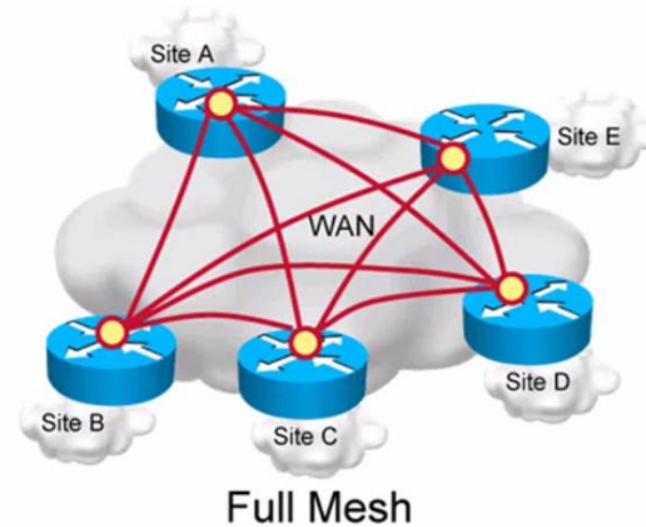
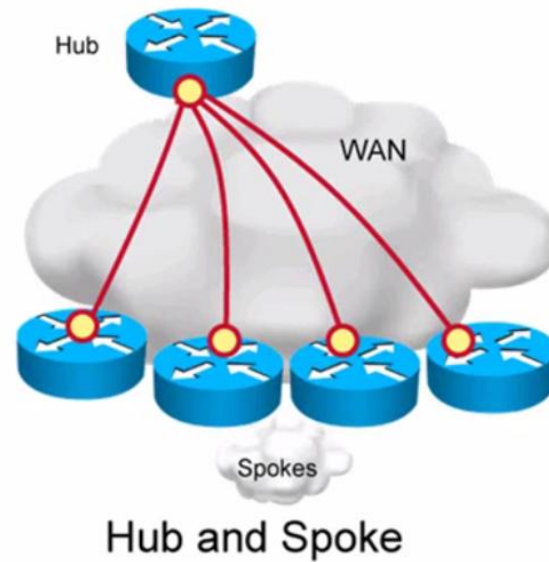
IPsec Framework

- Authentication
 - Device
 - User
- Encapsulation Method
- Data Encryption
- Packet Integrity
- Key Management
- Non-Repudiation
- Application and Protocol Support
- Address Management

Site-to-site VPN

- Provides secure communication between sites.
- The **physical** connectivity **does not affect** the capability of the VPN to successfully establish
 - Any **physical connection** the **condition** is **ping** between sites

Site-to-site VPN topology



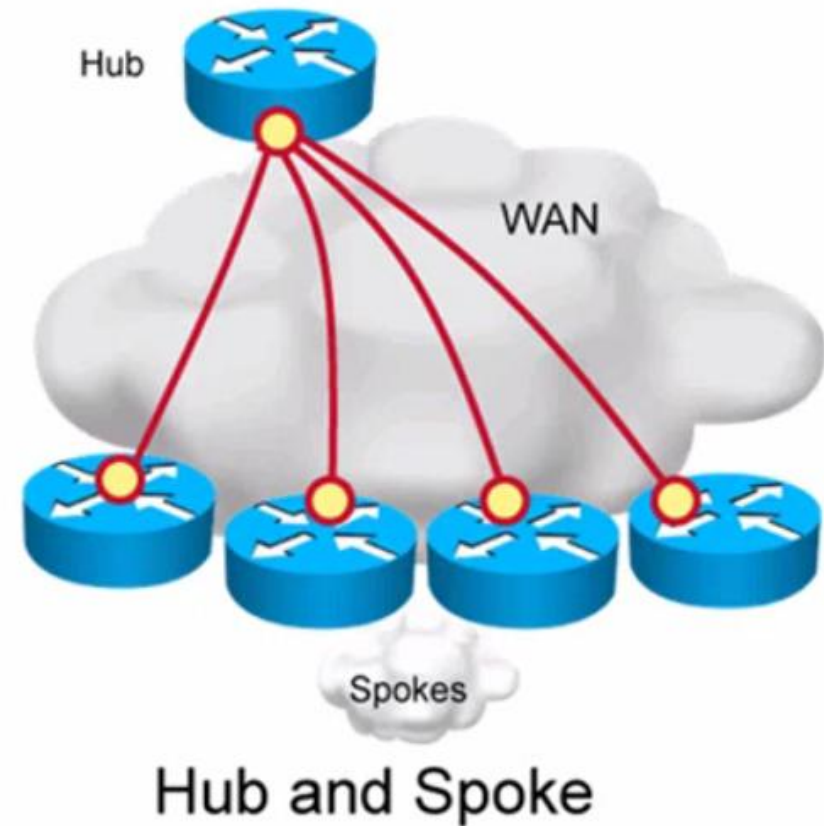
Individual P2P (point-to-point)

- Two sites connect to each other using secure VPN technologies



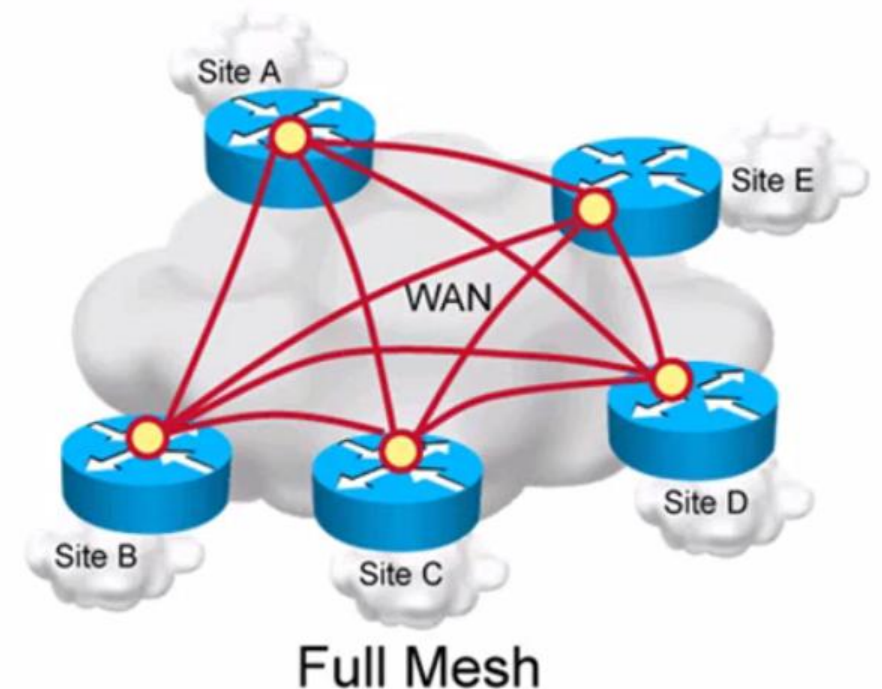
Hub-and-spoke network

- There is a **central site (hub)** and **all other sites (spokes) peer only with the hub**.
- **Connectivity between spokes** can be relayed **through the hub site**.
- Used when **multiple branches connect to the central branch**.
- Used when the **physical network infrastructure is already a hub-and-spoke design**



Meshed network

- **Fully meshed network**
 - **Multiple sites** that have a VPN connection to **each and every other site**.
 - This topology provides the **most optimal traffic flow**
- **Partially meshed network**
 - **Multiple sites** requiring **connectivity with other sites but not necessarily all**
 - **Each site** will have **multiple VPN connections to several other sites**

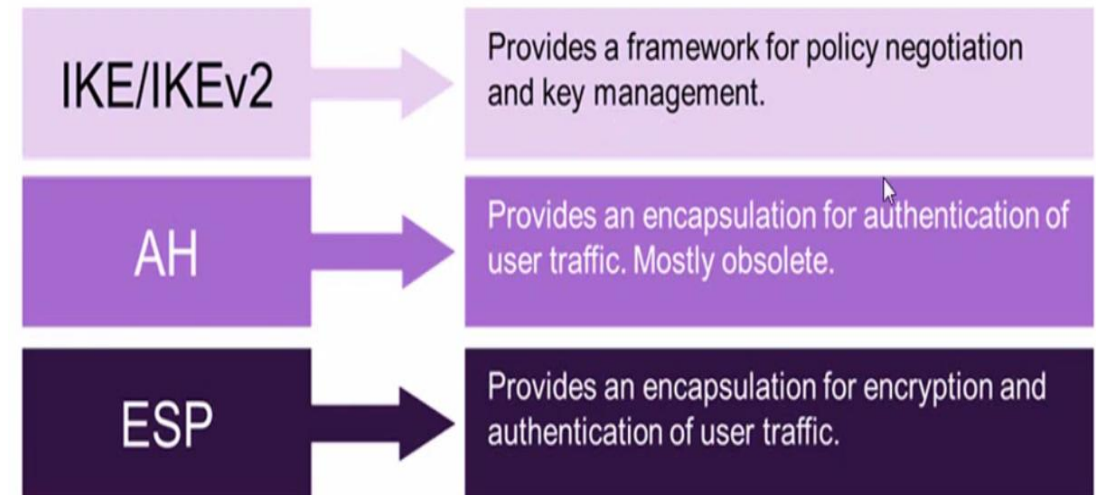


IPSec process

- **Initiation**: something has to **trigger** the creation of our tunnels.
 - For example when you configure **IPsec on a router**, you use an **access-list** to tell the **router what data to protect**.
 - When the **router receives something that matches the access-list**, it will **start the IKE process**. It's also possible to manually initiate the tunnel.
- **IKE phase 1**: we negotiate a **security association** to build the **IKE phase 1 tunnel (ISAKMP tunnel)**.
- **IKE phase 2**: within the **IKE phase 1 tunnel**, we build the **IKE phase 2 tunnel (IPsec tunnel)**.
- **Data transfer**: we **protect user data** by sending it through the **IKE phase 2 tunnel**.
- **Termination**: when there is **no user data to protect** then the **IPsec tunnel will be terminated after awhile**.

IKE overview

- IKE
 - Negotiation
 - Key management
- AH
 - Can do many parts of the IPsec objectives, **except encryption** of the data.
 - Work on port **51**
- ESP
 - Can do all the features of IPSec
 - Work on port **50**



Messages exchanged in Main Mode (Phase-1): (6 Messages)

Message No.	Initiator → Responder	Purpose
1	Initiator: sends IKE SA proposal	Proposes encryption, hashing, DH group, authentication method, lifetime
2	Responder: sends chosen IKE SA proposal	Selects matching parameters from proposal
3	Initiator: sends DH public key and random nonce (key exchange)	Begins Diffie-Hellman key exchange and introduces randomness
4	Responder: sends DH public key and random nonce	Completes key exchange to generate a shared secret
5	Initiator: sends identity and authentication info (encrypted)	Authenticates its identity (using PSK, RSA, or certificate)
6	Responder: sends identity and authentication info (encrypted)	Final identity/auth confirmation — secure channel is now established

After Message 6

- After Message 6:
- The ISAKMP Security Association (**ISAKMP SA**) is established.
- The **secure management tunnel** is now **active**, and the **devices are ready to negotiate IPSec SAs in Phase 2**.

Messages exchanged in Quick Mode (Phase-2): (3 Messages)

Message No.	Direction	Content	Purpose
1	Initiator → Responder	- Proposal for IPSec SA - Keying material (nonces) - Hash payload (auth check)	Proposes IPSec parameters (encryption, integrity, lifetime, PFS) and sends nonces for new key generation
2	Responder → Initiator	- Accepted IPSec SA - Keying material (nonces) - Hash payload	Responder select matching proposal and sends its nonce and hash
3	Initiator → Responder	- Hash payload	Final confirmation and authentication of SA establishment

After Quick Mode

- Two **IPSec Security Associations (SAs)** are established:
 - One for **inbound** traffic
 - One for **outbound** traffic
- IPSec can now begin encrypting **actual user data** using **ESP or AH**

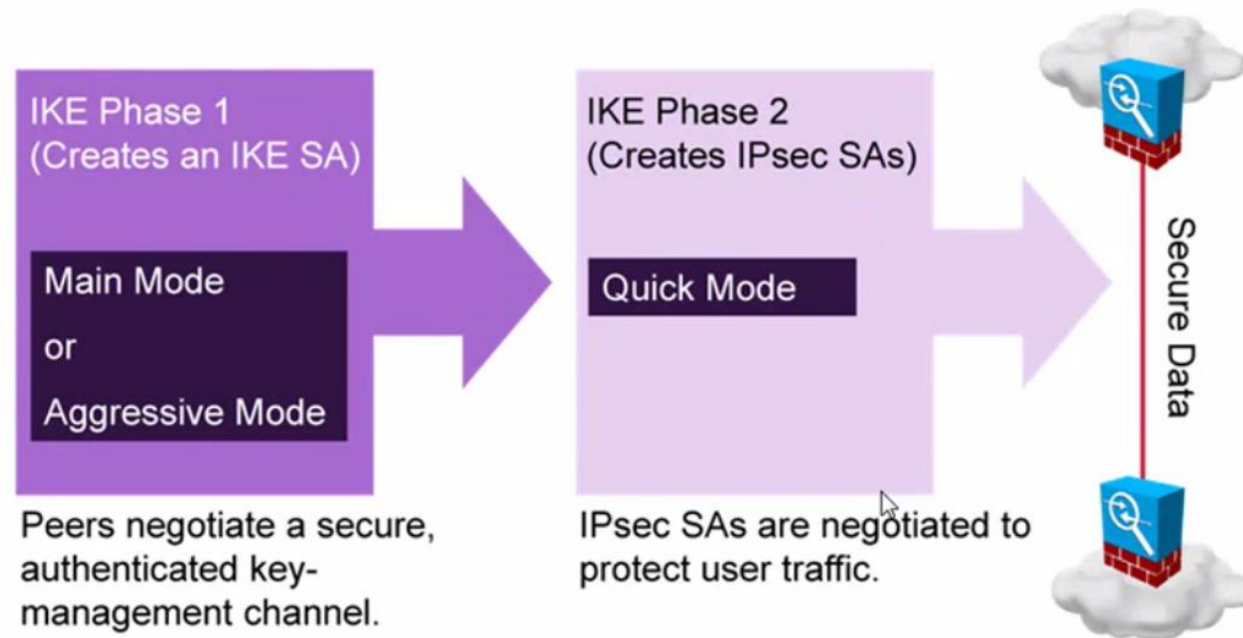
Internet Key Exchange (IKE) Protocol

- Used to **negotiate** and **establish** secured **site-to-site** or **remote access VPN tunnels**.
- **Layer 7** protocol
- IKE uses UDP, Port Number **500**.
- IKE is a framework
 - Internet Security Association and Key Management Protocol (ISAKMP)
 - ISAKMP defines **procedures** and **formats** to establish, negotiate, modify, and delete SA.
 - ISAKMP defines **payload** for exchanging key generation and authentication data.
 - ISAKMP **performs** peer authentication
 - two other **key management protocols**:
 - **Oakley protocol** uses the **Diffie-Hellman algorithm** to manage key exchanges across IPsec SAs.
 - **Secure Key Exchange Mechanism (SKEME)** used **RSA** to manage key exchanges

IKE v1, v2

Internet Key Exchange v1:

- Documented in RFC 2408
- Runs over UDP to destination port 500

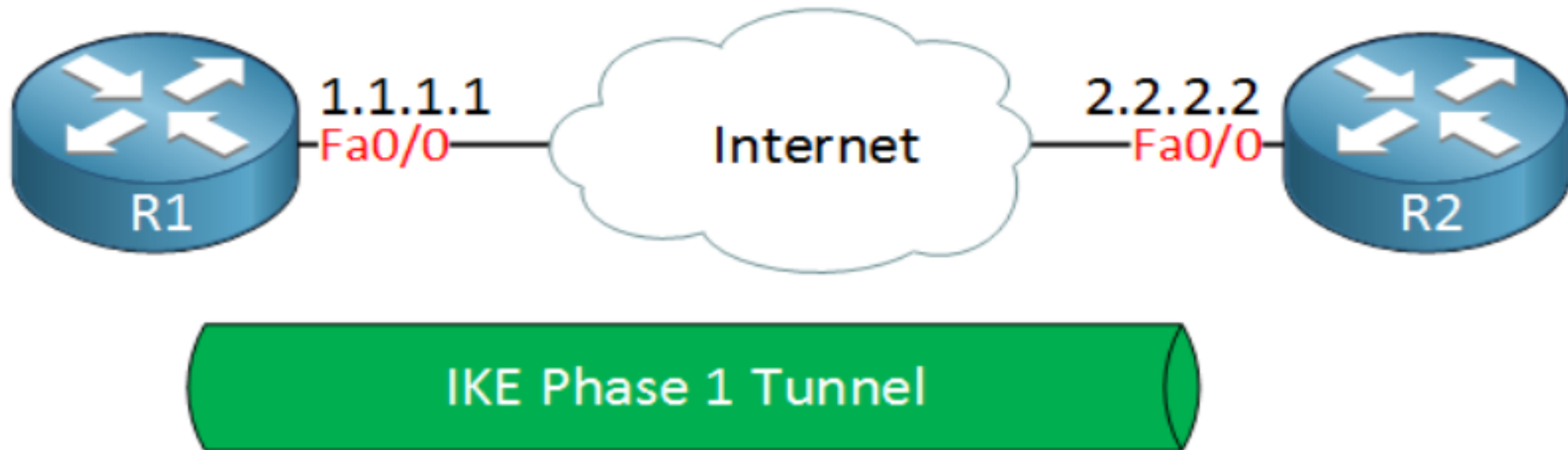


IKEv1 messages

- IKEv1 phase 1 using two different modes:
 - **Main mode**
 - **Aggressive mode**
- **Main mode** have 6 messages
- **aggressive mode** only uses 3 messages
- **IKEv1 phase 2** have 3 messages

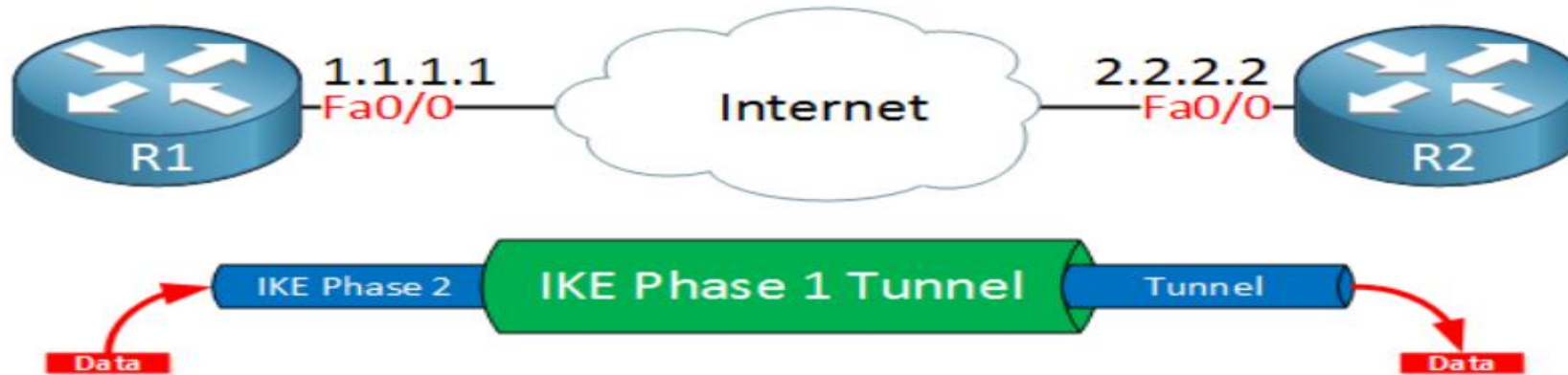
IKE phase 1 tunnel

- **The IKE phase 1 tunnel** is only used for **management IKE messages**
- create this **tunnel** as a *secure method to establish the second tunnel*
- the **IKE phase 2 tunnel** or **IPsec tunnel**
- for management traffic like keepalives



IKE phase 2 Tunnel

- use to protect user data
- This user data will be sent through the **IKE phase 2 tunnel** based on **IPSec algorithm**

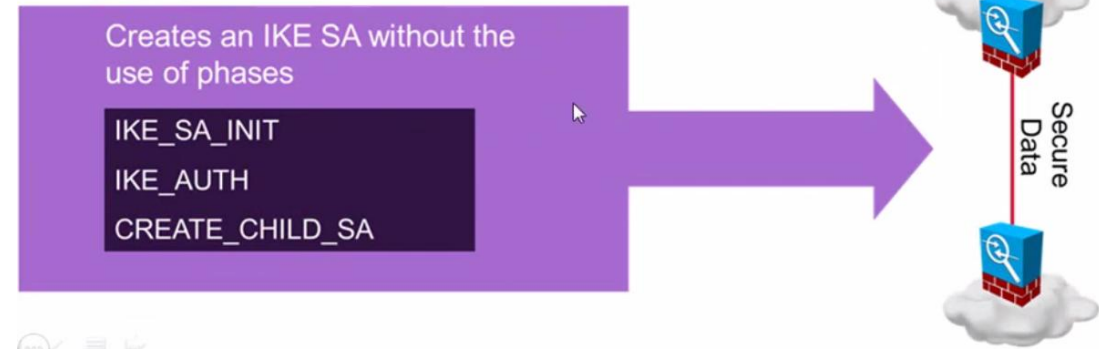


IKE v2

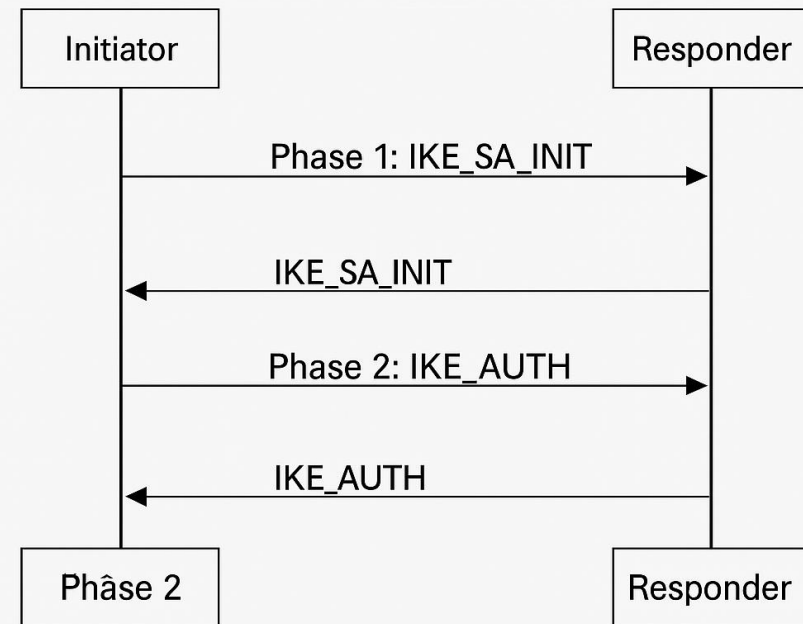
- Combine the 2 **stage** in **one stage**
- IKEv2 has **two phases** though
 - **phase 1** is called the IKE_SA_INIT
 - **Phase 2** is called IKE_AUTH

Internet Key Exchange v2:

- Documented in RFC 4306
- Runs over UDP to destination port 500
- There are two to five messages for basic exchange.
- Security Association (SA)
- IKEv2 creates the child SAs within the same negotiation, instead of using a phased approach.



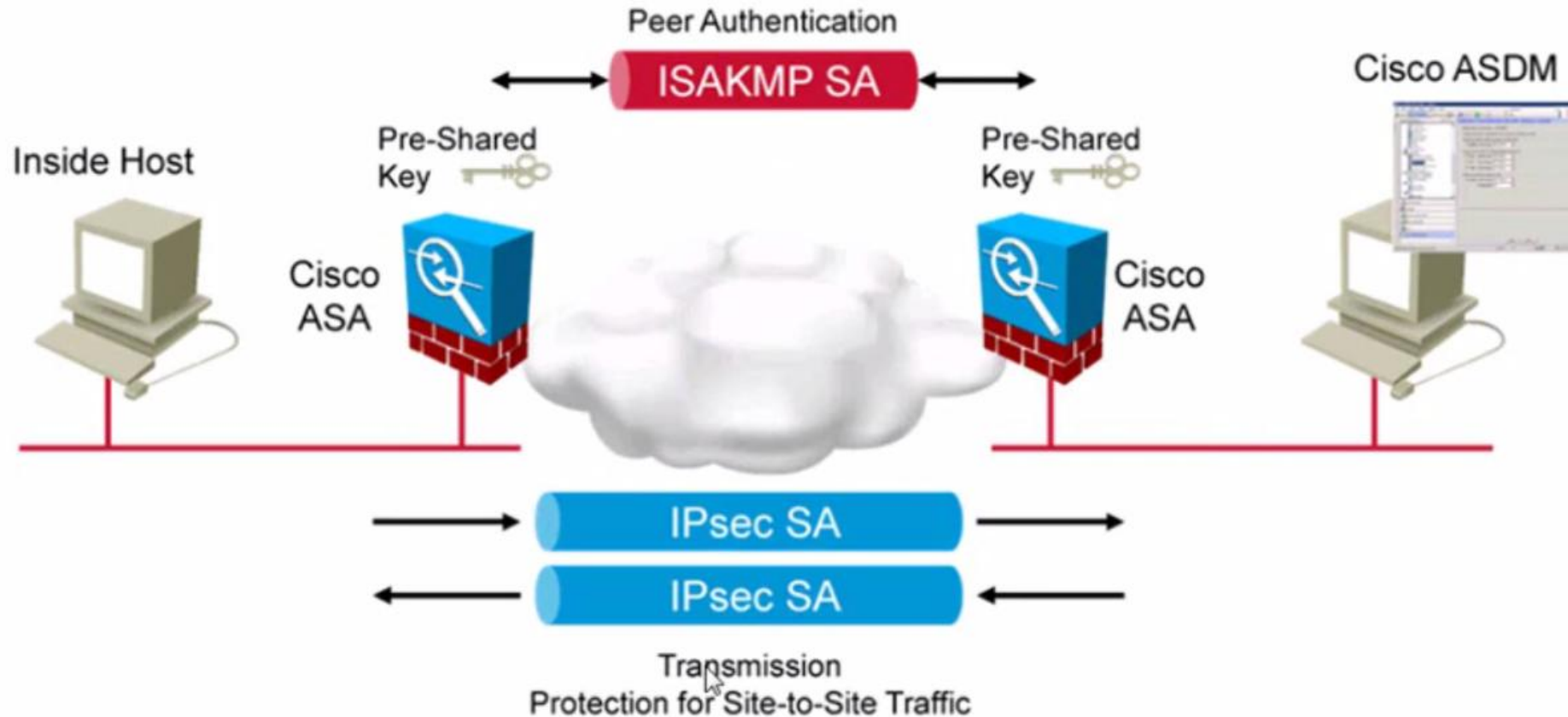
IKEv2



SA (Security Association)

- a **one-way** relationship between **sender & receiver** that affords security for **traffic flow**
- **defined by 3 parameters:**
 - **Security Parameters Index (SPI):** A bit string assigned to this SA
 - **IP Destination Address:** Only **unicast addresses** are allowed
 - **Security Protocol Identifier (ESP, AH)**
- has a number of other parameters
 - **seq no, AH & EH (ESP header) info, lifetime** etc.
- have a database of Security Associations

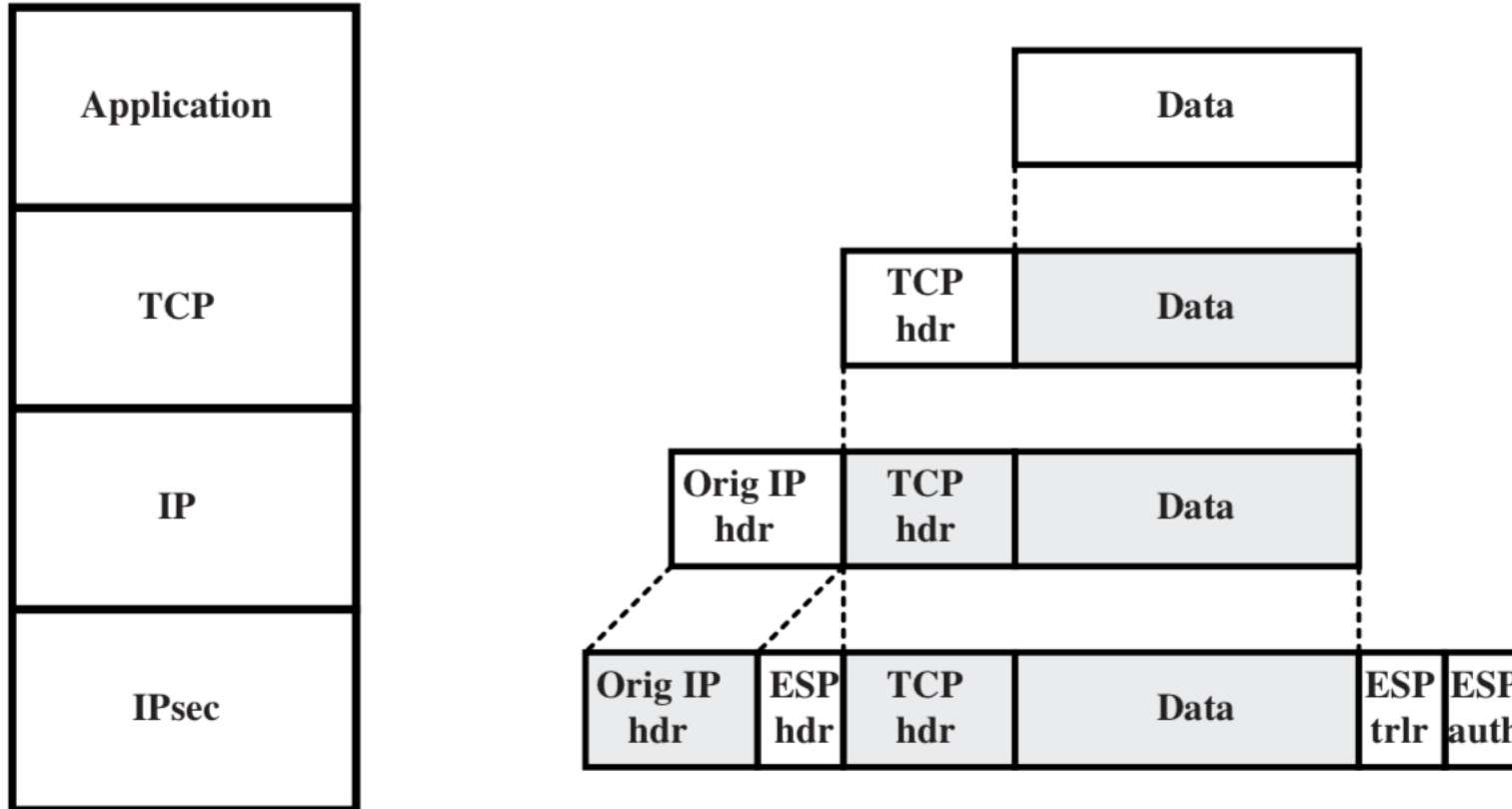
Point to point IPsec VPN



Transport and Tunnel Modes

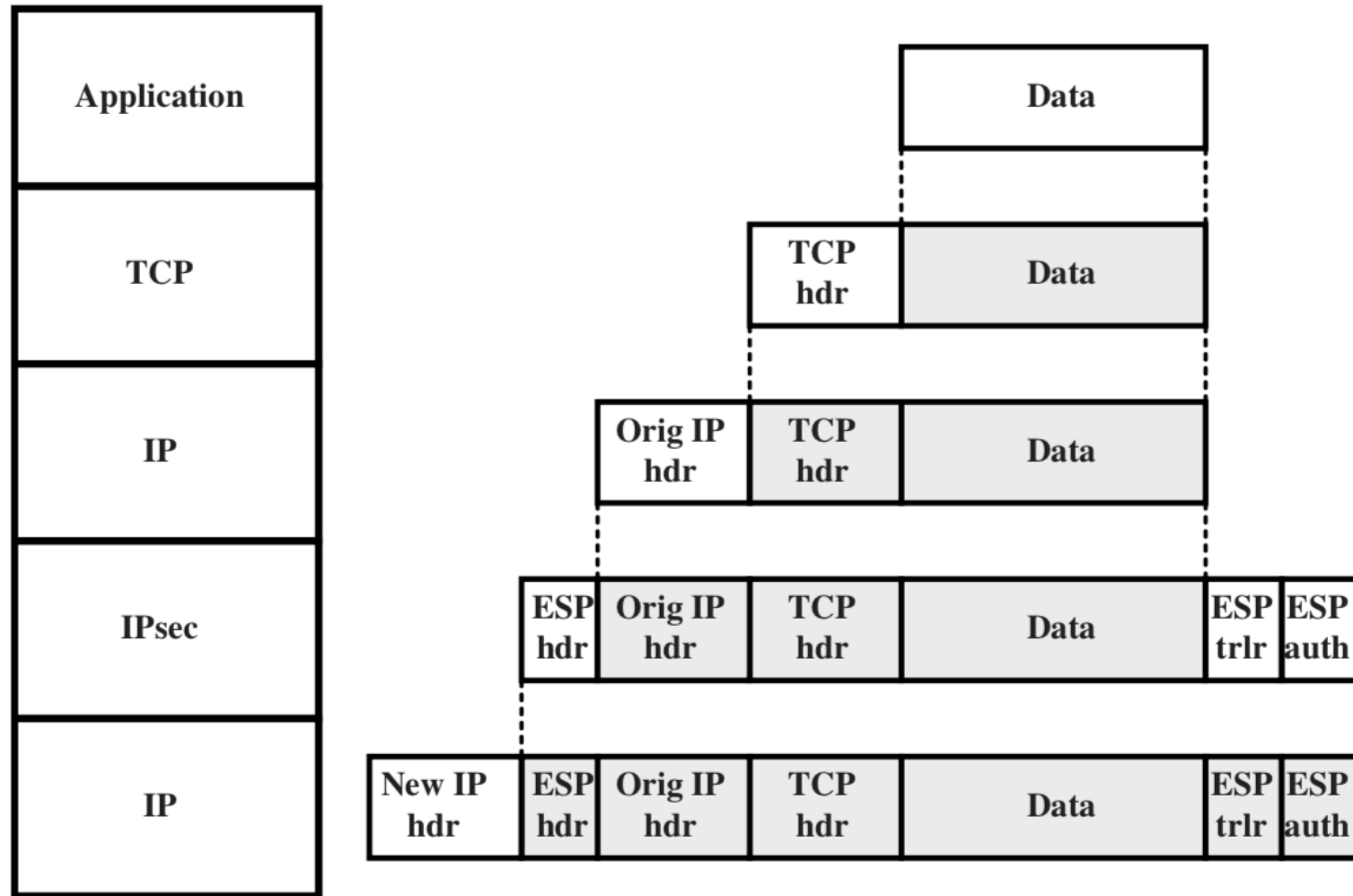
1. Both AH and ESP support two modes of use: transport and tunnel mode
2. Transport mode provides protection primarily for upper-layer protocols.
3. Tunnel mode provides protection to the entire IP packet.

IPSec header with transport mode



(a) Transport mode

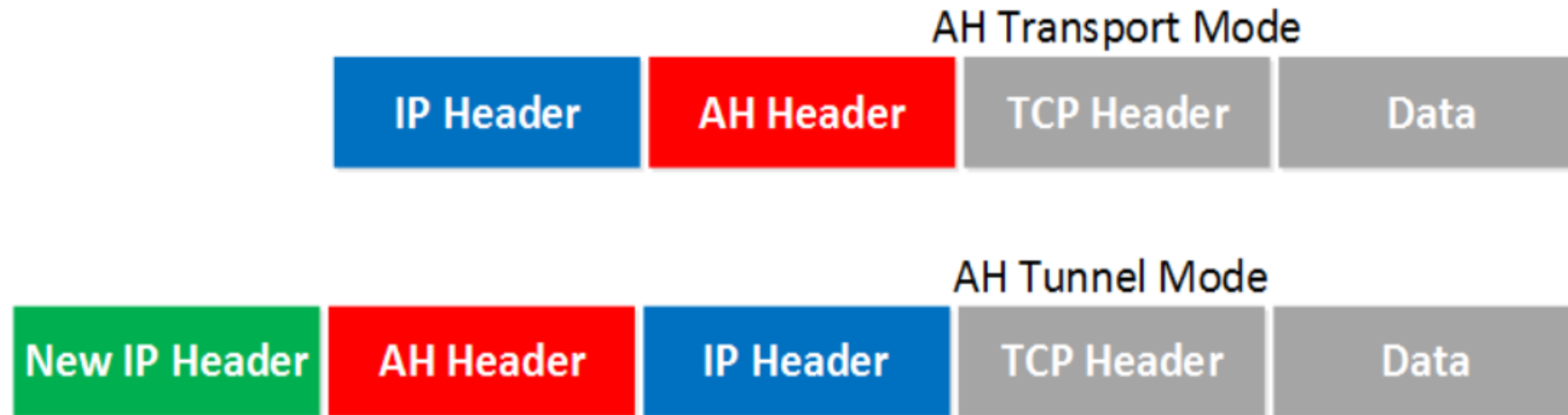
IPSec header with Tunnel Modes



(b) Tunnel mode

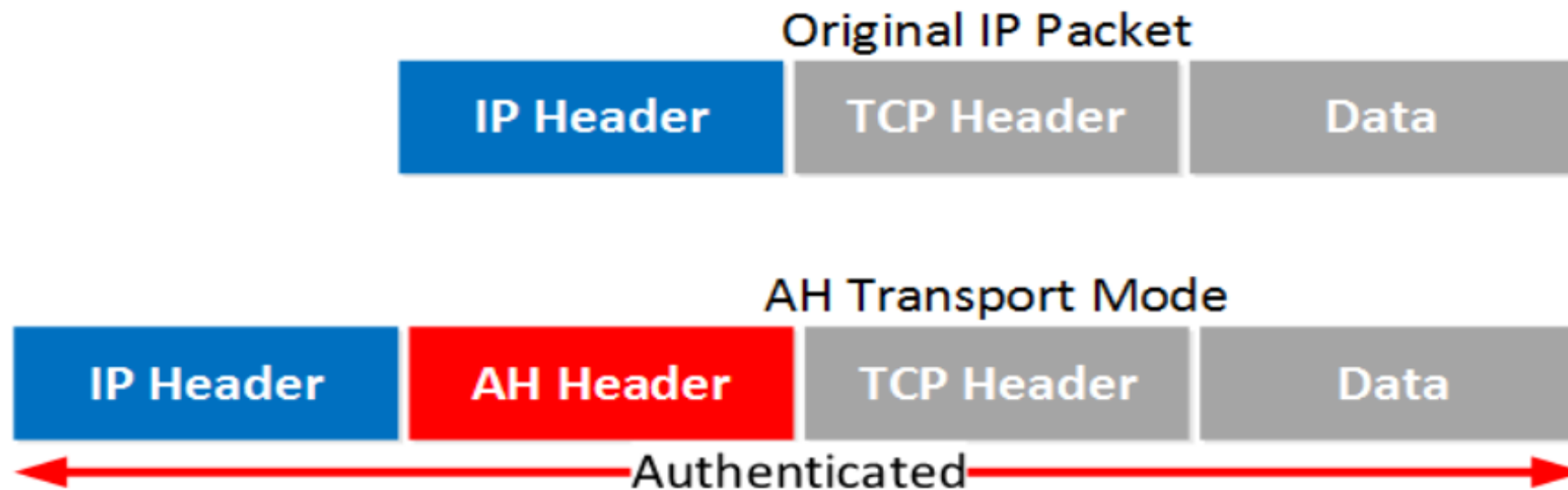
Transport and tunnel mode with AH

- transport mode will use the **original IP header**
- tunnel mode, use a new **IP header**



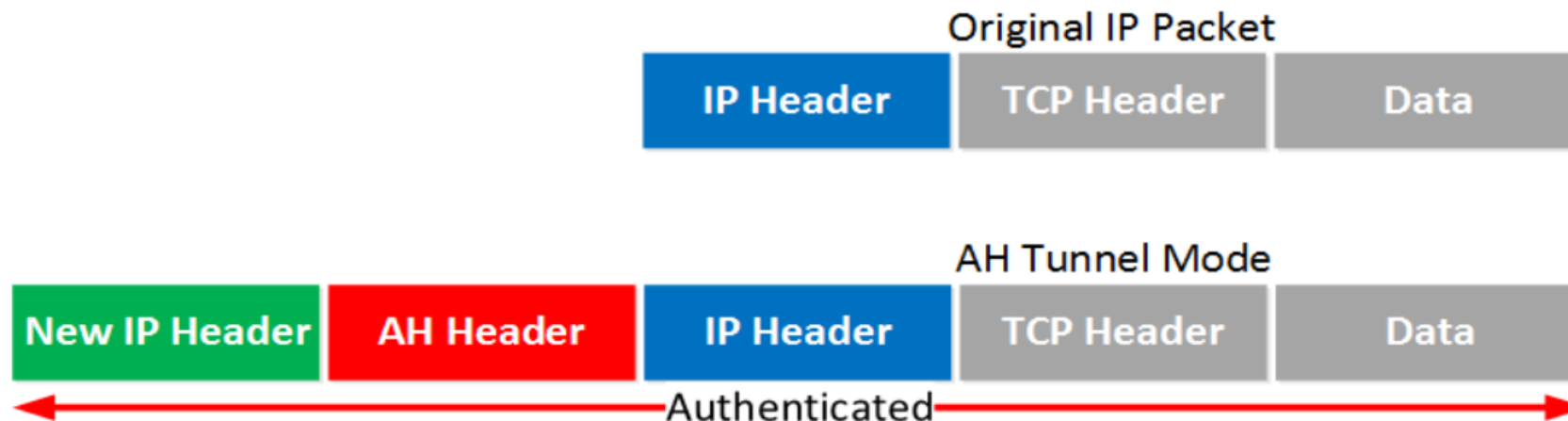
Transport Mode

- it just adds an **AH header** after the IP header



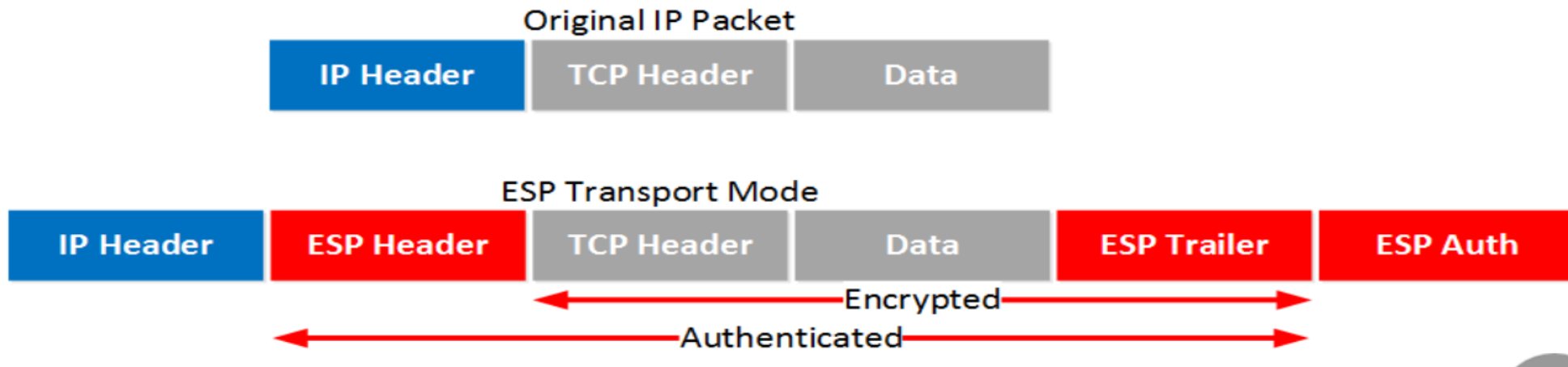
Tunnel Mode

- add a **new IP header** on top of the original IP packet
- when using **private IP addresses** and need to tunnel your traffic over the Internet
- It's **possible with AH** but it **doesn't offer encryption**
- **entire IP packet will be authenticated**



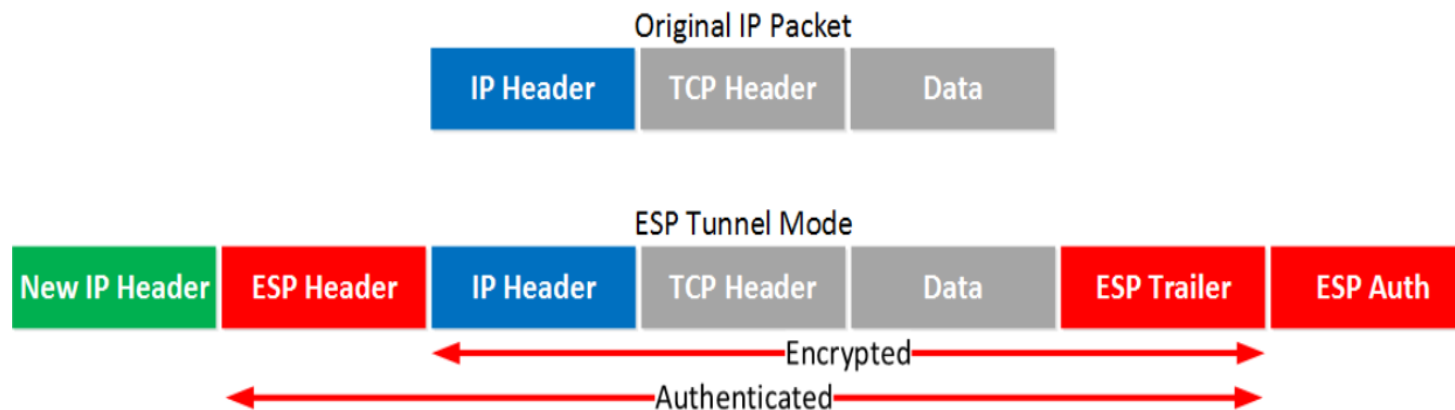
ESP Encapsulation with transport mode

- add an **ESP header and trailer**
- **transport layer (TCP for example) and payload** will be encrypted



ESP Encapsulation with tunnel mode

- new IP header which is useful for site-to-site VPNs
- similar to transport mode but we add a new header
- The original IP header is now also encrypted



Transport and Tunnel Modes

- For Transport mode using IPv4, the **ESP header** is **inserted** into the **IP packet immediately prior to** the transport-layer header (e.g., TCP, UDP), and an **ESP trailer** (Padding, Pad Length, and Next Header fields) is placed **after** the IP packet.
- If **authentication** is selected, the ESP Authentication Data field is **added after** the ESP trailer.
- The **entire** transport-level **segment** plus the **ESP trailer** are encrypted.
- **Authentication** covers **all of the ciphertext** plus the **ESP header**.

Transport and Tunnel Modes

- Tunnel mode ESP is used to **encrypt an entire IP packet**.
- For this mode, the ESP header is prefixed to **the packet and then the packet plus the ESP trailer is encrypted**.
- This method can be used to **counter traffic analysis**.
- Because the **IP header contains the destination address** and possibly source routing directives and hop-by-hop option information, it is not possible simply to transmit the encrypted IP packet prefixed by the ESP header.

Transport and Tunnel Modes

- Intermediate routers would be unable to process such a packet.
- Therefore, it is necessary to **encapsulate the entire block** (ESP header plus ciphertext plus Authentication Data, if present) with a **new IP header** that will contain **sufficient information for routing** but not for traffic analysis.
- Whereas the **transport mode** is suitable for **protecting** connections between **hosts** that support the **ESP** feature, the tunnel mode is useful in a configuration that includes a **firewall** or other sort of **security gateway** that protects a trusted network

Four combinations are possible tunnel mode

Host mode with AH	Host mode with ESP
Tunnel mode with AH	Tunnel mode with ESP

most common and
most important

Transport and Tunnel Modes

Table 19.1 Tunnel Mode and Transport Mode Functionality

	Transport Mode SA	Tunnel Mode SA
AH	Authenticates IP payload and selected portions of IP header and IPv6 extension headers.	Authenticates entire inner IP packet (inner header plus IP payload) plus selected portions of outer IP header and outer IPv6 extension headers.
ESP	Encrypts IP payload and any IPv6 extension headers following the ESP header.	Encrypts entire inner IP packet.
ESP with Authentication	Encrypts IP payload and any IPv6 extension headers following the ESP header. Authenticates IP payload but not IP header.	Encrypts entire inner IP packet. Authenticates inner IP packet.

IPSec Framework

- A Layer3 protocol provides confidentiality, data integrity, and authentication of the VPN

